

Technical Specification: Archivematica & AtoM

Integration on WSL2/Docker

Author: Marco Antonio Jiménez Gutiérrez

Contact: marco.jimenezgutierrez@ucr.ac.cr

Context: Developed for the Universidad de Costa Rica (UCR) — Archivo Universitario Rafael Obregón Loría (AUROL).

This document describes the architecture and configuration required to integrate Archivematica (1.x) and AtoM (2.x) running on the same Docker/WSL2 host, with specific focus on enabling DIP uploads from Archivematica to AtoM including thumbnail generation.

For installation steps, see the [Runbook](#).

Quick Reference

Service URLs and credentials

Service	URL	Username	Password
Archivematica Dashboard	http://localhost:62080	test	test
Archivematica Storage Service	http://localhost:62081	test	test
AtoM	http://localhost:63001	demo@example.com	demo

These are the default dev/hack environment credentials. Change them before any production use.

SFTP access (archivists uploading transfers)

Archivists deposit transfer packages into the system over SFTP using any standard client (WinSCP, FileZilla, or `sftp` CLI).

Field	Value
Protocol	SFTP
Host	IP of the Windows host (or localhost if connecting from the same machine)
Port	6222
Username	archivista
Password	Archivista2026
Remote folder	/home/archivista/transfer-source-sigedi

Files placed in that remote folder appear immediately in the Archivematica transfer source.

1. Architecture Overview

Both systems run in isolated Docker containers and do not share a filesystem by default. The integration uses two **host bind mounts** that bridge the gap without requiring network transfers between containers.



The two shared directories

Directory	Host path	Purpose
transfer-source	C:\ArchivematicaDrop\transfer-source	Archivists drop transfer packages here; Archivematica reads them as a transfer source
transfer-share	C:\ArchivematicaDrop\transfer-share	Archivematica writes DIPs here; AtoM reads them from its own container mount

The installation scripts inject both paths into `docker-compose.override.yml` automatically.

2. Integration Configuration

Even with the infrastructure automated by the install scripts, the following settings must be applied manually in the web interfaces to complete the integration.

A. Archivematica Dashboard — AtoM DIP Upload

Navigate to **Administration** → **AtoM DIP upload** in the Archivematica Dashboard (<http://localhost:62080>).

The screenshot shows the 'AtoM DIP upload' configuration page in the Archivematica Dashboard. The left sidebar lists various configuration options, with 'AtoM' selected under 'DIP upload'. The main content area contains the following settings:

- Upload URL ***: `http://host.docker.internal:63001/index.php`. Description: URL where the AtoM index.php frontend lives, SWORD services path will be appended.
- Login email ***: `demo@example.com`. Description: E-mail account used to log into AtoM.
- Login password ***: `demo`. Description: Password used to log into AtoM.
- AtoM version ***: `2.x`.
- Rsync target**: `/home/transfer-share-from-archivematica`. Description: The DIP can be sent with Rsync to a remote host before is deposited in AtoM. This is the destination value passed to Rsync (see man 1 rsync). For example: `foobar.com:~/dips/`.
- Rsync command**: (empty field). Description: If `--rsync-target` is used, you can use this argument to specify the remote shell manually. For example: `ssh -p 22222 -l user`.
- Debug mode**: `No`. Description: Show additional details.
- REST API key**: `5d7ff26a4910de84`. Description: Used in metadata-only DIP upload.

A 'Save' button is located at the bottom of the form.

Setting	Value	Notes
Upload URL	<code>http://host.docker.internal:63001/index.php</code>	Routes from the Archivematica container back through the host to AtoM's port. Using <code>localhost</code> here would resolve inside the Archivematica container and fail.
Login email	<code>demo@example.com</code>	AtoM administrator account
Login password	<code>demo</code>	AtoM administrator password
AtoM version	<code>2.x</code>	
Rsync target	<code>/home/transfer-share-from-archivematica</code>	Path <i>inside the Archivematica container</i> where the DIP is placed
REST API key	(see below)	Required for authorization and metadata mapping

Getting the AtoM REST API key: In AtoM, go to **Admin** → **Users**, edit the administrator user, and copy the value from the **REST API key** field. If it is empty, click **Generate** to create one.

B. Archivemata Storage Service — Verify pipeline locations

Navigate to **Locations** in the Storage Service (<http://localhost:62081/locations/>) and confirm that the pipeline's transfer source and AIP storage locations are active and pointing to the expected paths.

Archivemata Dashboard - Ad...Archivemata Storage ServiceVAS-5950-2024.pdf - Demo siteRecibidos (5) - marjingu@gmail.com

localhost:62081/locations/

Archivemata Storage ServiceHomePipelinesSpacesLocationsPackagesAdministrationLog out

All Locations

Locations have a purpose, such as AIP storage or transfer source, and are where packages are stored. Each Location is associated with a pipeline and can only be accessed by that pipeline.

Show10entries

Purpose	Pipeline	Path	Description	Space	UUID	Usage	Enabl
AIP Recovery	Archivemata on d50a43b17886	/var/archivemata/storage_service/recover	Default AIP recovery	badfa217-0...	9ae50212-0c2f-4764-8cef-db3c0dc61094	0B / unlimited	Enabl
AIP Storage	Archivemata on d50a43b17886	/var/archivemata/sharedDirectory/www/AIPsStore	Store AIP in standard Archivemata Directory	badfa217-0...	e9c100b2-fa52-4c62-bdc9cdbe876f896	188098B / unlimited	Enabl
Currently Processing	Archivemata on d50a43b17886	/var/archivemata/sharedDirectory	None	badfa217-0...	f66a4907-37b0-4fd2-a2ea-105b5a58a3ee	0B / unlimited	Enabl
DIP Storage	Archivemata on d50a43b17886	/var/archivemata/sharedDirectory/www/DIPsStore	Store DIP in standard Archivemata Directory	badfa217-0...	e34070b3-9b53-439b-9c5a-b72fa2bcb7c2	936052B / unlimited	Enabl
Storage Service Internal Processing	No pipelines	/var/archivemata/storage_service	For storage service internal usage.	badfa217-0...	0112af7d-5398-48ed-a218-870873040584	0B / unlimited	Enabl
Transfer Backlog	Archivemata on d50a43b17886	/var/archivemata/sharedDirectory/www/AIPsStore/transferBacklog	Default transfer backlog	badfa217-0...	c88d8d2b-32aa-4d9d-9d05-e6785ef26a8f	0B / unlimited	Enabl
Transfer Source	Archivemata on d50a43b17886	/home	Default transfer source	badfa217-0...	c3b1adfc-9802-498e-a181-ec6479f854dd	0B / unlimited	Enabl

Showing 1 to 7 of 7 entries

C. AtoM Global Settings — SWORD deposit directory

Navigate to **Admin** → **Settings** → **Global** in AtoM (<http://localhost:63001>).

Archivematica Dashboard - Ad... Archivematica Storage Service Global setting - Demo site Recibidos (5) - marjingu@gmail.com

localhost:63001/index.php/settings/global

Clipboard

DIP upload

Default page elements

Default template

Diacritics

Digital object derivatives

Finding Aid

Global

Header customizations

I18n languages

Identifiers

Inventory

Markdown

Permissions

Privacy Notification

Security

Site information

Storage service

Treeview

Uploads

User interface labels

Web analytics

Application version

2.10.1 - 197

The current version of the application

Check for updates

☐ No

☒ Yes

Enable automatic update notification

Search and browse

Presentation

Multi-repository

Permalinks

System

Integrations

Google Maps Javascript API key (for displaying dynamic maps)

SWORD deposit directory

/home/transfer-share-from-atom

Save

Setting	Value	Notes
SWORD deposit directory	/home/transfer-share-from-atom	Path <i>inside the AtoM container</i> where it looks for incoming DIPs

If the field is missing from the UI: The setting can be hardcoded in `apps/qubit/config/app.yml` as `sword_deposit_dir: /home/transfer-share-from-atom`. After editing, clear the Symfony cache (`php symfony cc`) and restart `atom_worker`.

3. End-to-End Workflow

Use this sequence to validate the integration from start to finish.

- 1. Deposit a transfer** — Copy a folder containing PDFs or images to the transfer source:
 - Via SFTP: connect with the credentials above and upload to `/home/archivista/transfer-source-sigedi`
 - Directly on the host: copy to `C:\ArchivematicaDrop\transfer-source`
- 2. Start a transfer in Archivematica** — Dashboard → **Transfer** tab → Standard transfer → Browse → select the folder → Start transfer.
- 3. Process through the pipeline** — Approve each micro-service step until the **Upload DIP** micro-service appears. Select the configured AtoM upload target.

4. **Monitor logs** — If the upload fails or stalls, watch the relevant container logs side by side:

```
# Archivematica MCP Client (sends the DIP)
cd ~/src/archivematica/hack
docker compose logs -f archivematica-mcp-client

# AtoM Worker (receives DIP, generates thumbnails)
cd ~/src/atom
export COMPOSE_FILE="docker/docker-compose.dev.yml:docker/docker-
compose.override.yml"
docker compose logs -f atom_worker
```

5. **Verify in AtoM** — Navigate to <http://localhost:63001>. The descriptive record should be present and the digital object viewer should show the uploaded files with generated thumbnails.

4. Remote Setup: SSH/Rsync

The shared bind-mount architecture works only when both systems run on the same physical host. For distributed deployments (Archivematica and AtoM on separate servers), the DIP must travel over the network via Rsync/SSH.

Archivematica side (sender)

1. Generate an SSH key pair for the `archivematica-mcp-client` container user.
2. Set **Rsync target** to a remote SSH path: `user@atom-server.example.com:/path/to/deposit/`
3. If using a non-standard port or key: set **Rsync command** to `ssh -i /home/archivematica/.ssh/id_rsa -p 2222`

AtoM side (receiver)

1. Ensure `sshd` is running and reachable from the Archivematica server's IP.
2. Add the Archivematica public key to `~/.ssh/authorized_keys` of the receiving user.
3. Ensure the receiving user has write permission to the SWORD deposit directory. If the SSH user differs from the PHP-FPM user (`www-data`), use a shared group with `chmod g+s` and `775` permissions so the AtoM worker can still read the files.